

BRM Unit 9 Exam Notes

Unit 9 In One Line

Unit 9 is about **measuring variables properly and checking whether the measurement tool is reliable and valid before analysis.**

In 2 words: **Measurement Quality**

Why Unit 9 Matters

Unit 9 is asked very often in PYQs. The most repeated topics are:

1. Reliability
2. Validity
3. Goodness of measures
4. Rating scales
5. Measurement error
6. Operational definitions
7. Basic statistical technique selection

The exam usually asks theory plus examples. In 2026, Unit 9 came strongly through reliability, validity, measurement error, and choosing a statistical technique.

1. What Is Measurement?

Measurement means assigning numbers or symbols to characteristics of objects according to rules.

Simple meaning:

Measurement = converting concepts into numbers or categories.

Examples:

- Gender: Male = 1, Female = 2
- Satisfaction: 1 = Very dissatisfied, 5 = Very satisfied
- Income: actual monthly income
- Customer loyalty: measured using repeat purchase, recommendation, and preference

Exam definition:

Measurement is the assignment of numbers or symbols to characteristics of objects according to pre-defined rules.

2. Why Measurement Is Important

In research, we cannot test hypotheses unless variables are measured.

Example:

If the hypothesis is:

Higher customer trust leads to higher customer loyalty.

Then we must measure:

- Customer trust
- Customer loyalty

Without measurement, the hypothesis cannot be tested.

3. Objective And Subjective Variables

Variables can be:

1. Objective variables
2. Subjective variables

4. Objective Variables

Objective variables are easy to measure directly.

Simple meaning:

Can be counted or physically measured.

Examples:

- Age
- Salary
- Number of products produced
- Absenteeism
- Years of experience
- Number of rejects per month

5. Subjective Variables

Subjective variables are abstract and cannot be measured directly.

Simple meaning:

Feelings, opinions, attitudes, perceptions.

Examples:

- Customer trust
- Job satisfaction

- Employee engagement
- Organizational climate
- Service quality
- Brand loyalty

For subjective variables, we need operational definitions and rating scales.

6. Operational Definition

Operational definition means defining an abstract concept in measurable terms.

Simple meaning:

Operational definition = how exactly you will measure a concept.

Example:

Customer trust is abstract. You cannot see trust directly.

So you measure it through:

- Belief that the brand keeps promises
- Feeling that the brand is honest
- Confidence in product quality
- Willingness to share personal information
- Belief that the brand cares about customers

Exam definition:

Operational definition is a statement of the specific dimensions and measurable elements through which a concept is measured.

7. Dimensions And Elements

To operationalize a concept, break it into:

1. Dimensions
2. Elements/items

Dimension:

Broad part of the concept.

Element:

Specific measurable question/behavior.

Example: Consumer Trust

	Dimension		Measurable item	
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Honesty	The brand provides truthful information.
Reliability	The brand delivers what it promises.
Security	I feel safe sharing information with the brand.
Customer care	The brand cares about customer problems.

8. Operational Definition Example: Employee Engagement

Employee engagement can be measured through:

Dimensions:

1. Emotional involvement
2. Commitment to organization
3. Motivation to perform
4. Willingness to go beyond basic duties

Possible questions:

- I feel emotionally connected to my organization.
- I am willing to put extra effort into my work.
- I feel motivated to achieve organizational goals.
- I would recommend this organization as a good place to work.

Responses can be measured on a 5-point Likert scale:

1 = Strongly disagree, 5 = Strongly agree

9. What Operational Definition Is Not

Operational definition should measure the concept itself, not its causes or results.

Example:

Achievement motivation may lead to good performance.

But performance is not the same as achievement motivation.

Wrong:

Measuring motivation only through salary or success.

Correct:

Measuring motivation through behavior such as persistence, preference for challenging work, feedback-seeking, and goal focus.

Shortcut:

Do not measure the result of a concept. Measure the behavior that shows the concept.

10. Data Analysis Workflow

Unit 9 starts with the idea that raw data must be cleaned before analysis.

Steps:

1. Collect raw data
2. Data refinement
3. Data pre-processing
4. Reliability testing
5. Validity testing
6. Hypothesis testing

Simple meaning:

Before doing analysis, clean the data and check whether the tool is good.

11. Data Refinement

Data refinement means cleaning raw data before analysis.

Simple meaning:

Data refinement = data cleanup.

It includes:

1. Checking incomplete answers
2. Checking inconsistent answers
3. Handling missing values
4. Handling outliers

Example:

If a respondent leaves 25 percent or more of the questionnaire blank, the response may be discarded.

12. Data Editing

Data editing means checking whether responses are complete and consistent.

Examples:

- A person says marital status is single but writes spouse occupation.
- A respondent skips many important questions.
- A person gives impossible age like 200 years.

13. Missing Values

Missing values are unanswered questions in the data.

How to handle:

- If too many answers are missing, discard the questionnaire.
- If only a few are missing, use mean substitution or ignore that case for that analysis.

Mean substitution:

Replacing missing value with the average response.

14. Outliers

Outliers are extreme values that are very different from the rest.

Example:

In a middle-class income survey, one person enters monthly income as Rs. 10 crore.

Why outliers matter:

They can distort averages and analysis.

15. Data Pre-Processing

Data pre-processing prepares cleaned data for analysis.

It includes:

1. Data coding
2. Data entry
3. Data transformation

16. Data Coding

Data coding means assigning numbers to responses.

Examples:

- Male = 1, Female = 2
- Yes = 1, No = 0
- Strongly disagree = 1, Strongly agree = 5

Codebook:

A codebook records what each number means.

17. Data Entry

Data entry means entering coded responses into Excel, SPSS, or any analysis software.

Example:

Each row = one respondent.

Each column = one question/variable.

18. Data Transformation

Data transformation means changing data into a useful form for analysis.

Two common types:

1. Reverse coding
2. Aggregation

19. Reverse Coding

Reverse coding is used for negatively worded questions.

Example:

Question:

I do not trust this brand.

If 5 = strongly agree, then high score means low trust. So it must be reversed.

On a 5-point scale:

- 5 becomes 1
- 4 becomes 2
- 3 stays 3
- 2 becomes 4
- 1 becomes 5

Why:

So all questions move in the same direction.

20. Aggregation

Aggregation means combining several items to create one variable score.

Example:

Customer satisfaction is measured using 5 questions. The average of those 5 answers becomes the customer satisfaction score.

21. Reliability

Reliability means consistency of a measurement tool.

Simple meaning:

Reliability = same result again and again.

If a weighing scale shows 60 kg today, 60 kg tomorrow, and 60 kg next week for the same person, it is reliable.

Research example:

If a customer satisfaction questionnaire gives similar results when used repeatedly in similar conditions, it is reliable.

Exam definition:

Reliability refers to the extent to which a measurement instrument produces consistent and stable results.

22. Why Reliability Is Important

Reliability is important because inconsistent tools produce wrong conclusions.

Example:

If the same employee gives very different engagement scores without any real change, the questionnaire is unreliable.

Importance:

- Ensures consistency
- Reduces random error
- Increases trust in research findings
- Helps compare results
- Supports better decision-making

23. Methods To Assess Reliability

Important methods:

1. Test-retest reliability
2. Parallel-form reliability
3. Internal consistency reliability
4. Split-half reliability
5. Cronbach's Alpha

24. Test-Retest Reliability

The same instrument is given to the same respondents at two different times.

If results are similar, the tool is reliable.

Example:

A job satisfaction survey is given to employees in January and again in February. If scores are similar and no major workplace change occurred, the tool has test-retest reliability.

25. Parallel-Form Reliability

Two equivalent versions of the same test are given to the same respondents.

If both forms give similar results, reliability is high.

Example:

Two different but similar customer trust questionnaires give similar scores.

26. Internal Consistency Reliability

Internal consistency checks whether items measuring the same concept give consistent answers.

Simple meaning:

All questions measuring the same thing should agree with each other.

Example:

If five questions measure customer satisfaction, responses to those questions should be related.

27. Cronbach's Alpha

Cronbach's Alpha checks internal consistency.

Simple meaning:

It tells whether items in a scale hang together.

Benchmark:

- Below 0.60 = poor
- Around 0.70 = acceptable
- 0.80 and above = good/excellent

Exam sentence:

Cronbach's Alpha is commonly used to measure internal consistency reliability. A value of 0.70 or above is generally considered acceptable.

28. Split-Half Reliability

The questionnaire items are divided into two halves and scores from both halves are compared.

If both halves give similar results, reliability is high.

Example:

A 10-item satisfaction scale is split into odd-numbered and even-numbered items. If both sets give similar satisfaction scores, the scale is reliable.

29. Validity

Validity means accuracy of the measurement tool.

Simple meaning:

Validity = measuring what you are supposed to measure.

Example:

If a test is supposed to measure customer trust, it should actually measure trust, not only customer satisfaction or brand awareness.

Exam definition:

Validity refers to the extent to which a research instrument measures what it is intended to measure.

30. Reliability vs Validity

Reliability	Validity
Consistency	Accuracy
Same result repeatedly	Correct concept measured
Can be reliable but not valid	Valid tool is generally reliable
Example: scale gives same wrong weight daily	Example: scale gives correct weight

Shortcut:

- **Reliability = consistent**
- **Validity = correct**

31. Important Statement

Question:

"A valid instrument is always reliable, but a reliable instrument may not always be valid." Explain.

Answer:

If an instrument is valid, it measures the correct concept accurately, so it must also be reasonably consistent. Therefore, validity includes reliability.

However, a tool may be reliable but not valid. For example, a weighing scale may consistently show 5 kg more than actual weight. It is reliable because it gives the same result repeatedly, but it is not valid because the result is inaccurate.

Similarly, a questionnaire may consistently measure customer satisfaction, but if the research aim is to measure customer trust, then it is reliable but not valid.

32. Types Of Validity

Important types:

1. Content validity
2. Criterion validity
3. Construct validity
4. Convergent validity
5. Discriminant validity

33. Content Validity

Content validity checks whether the instrument covers the full concept.

Simple meaning:

Did we cover all important parts?

Example:

Customer satisfaction questionnaire should include product quality, price, service, delivery, and complaint handling.

If it only asks about price, content validity is weak.

34. Criterion Validity

Criterion validity compares the instrument with a known standard or outcome.

Simple meaning:

Does it match a real-world result or accepted standard?

Example:

A sales aptitude test has criterion validity if people scoring high on it actually perform better in sales jobs.

35. Construct Validity

Construct validity checks whether the instrument truly measures the theoretical concept.

Simple meaning:

Does the scale behave as theory says it should?

Example:

Items meant to measure customer trust should group together and represent trust, not unrelated concepts.

Factor analysis can help assess construct validity.

36. Convergent Validity

Convergent validity means two measures of the same concept should be highly related.

Simple meaning:

Same concept, similar result.

Example:

Two different scales measuring customer loyalty should show high correlation.

37. Discriminant Validity

Discriminant validity means measures of different concepts should not be too highly related.

Simple meaning:

Different concepts should stay different.

Example:

Customer trust and product variety are different concepts, so their scales should not behave as if they are exactly the same.

38. Goodness Of Measures

Goodness of measures means checking whether the measurement tool is both reliable and valid.

Simple meaning:

Good measure = consistent + correct.

A good measurement tool should:

1. Measure the correct concept
2. Give consistent results
3. Reduce measurement error
4. Help produce trustworthy findings

39. Rating Scales

Rating scales are used to measure attitudes, opinions, satisfaction, and perceptions.

Simple meaning:

Rating scale = respondents select a level or score.

40. Likert Scale

Respondents show agreement or disagreement.

Example:

The brand keeps its promises.

1. Strongly disagree
2. Disagree
3. Neutral
4. Agree
5. Strongly agree

Use:

Attitude, satisfaction, trust, engagement.

41. Semantic Differential Scale

Respondents rate between two opposite words.

Example:

Bank service is:

Slow 1 2 3 4 5 Fast

Unreliable 1 2 3 4 5 Reliable

Use:

Brand image, service perception, product image.

42. Numerical Scale

Respondents give a number rating.

Example:

Rate your satisfaction from 1 to 10.

1 = Very dissatisfied, 10 = Very satisfied

43. Ranking Scale

Respondents rank options in order of preference.

Example:

Rank the following factors while choosing a phone:

1. Price
2. Camera
3. Battery
4. Brand

Use:

Preference ordering.

44. Dichotomous Scale

Respondents choose between two options.

Example:

Have you used mobile banking?

Yes / No

Use:

Simple yes/no questions.

45. Measurement Error

Measurement error means the difference between the true value and the measured value.

Simple meaning:

Measurement error = wrong or inaccurate answer caused by poor measurement.

In questionnaire design, errors happen when questions are unclear, biased, confusing, or poorly worded.

46. Types Of Measurement Errors In Questionnaire

Important types for exam:

1. Ambiguous question
2. Leading/loading question
3. Double-barreled question
4. Social desirability bias
5. Recall error
6. Scale error

47. Ambiguous Question

Meaning:

The question is unclear and can be understood in many ways.

Bad example:

Is our service good?

Problem:

"Good" is vague.

Improved question:

How satisfied are you with the speed of our customer service response?

Scale: 1 = Very dissatisfied, 5 = Very satisfied

48. Leading Question

Meaning:

The question pushes the respondent toward a particular answer.

Bad example:

How satisfied are you with our excellent customer service?

Problem:

The word "excellent" influences the answer.

Improved question:

How satisfied are you with our customer service?

49. Double-Barreled Question

Meaning:

One question asks about two things at once.

Bad example:

Are you satisfied with our price and delivery speed?

Problem:

Customer may like price but dislike delivery.

Improved questions:

1. How satisfied are you with our pricing?
2. How satisfied are you with our delivery speed?

50. Social Desirability Bias

Meaning:

Respondents give answers that make them look good instead of truthful answers.

Bad example:

Do you always buy eco-friendly products?

Improved question:

How often do you buy eco-friendly products?

Options:

Never, Rarely, Sometimes, Often, Always

51. Recall Error

Meaning:

Respondents cannot accurately remember past events.

Bad example:

How many times did you contact customer support in the last 5 years?

Improved question:

How many times did you contact customer support in the last 3 months?

52. Scale Error

Meaning:

The response scale is poorly designed or unbalanced.

Bad example:

How satisfied are you?

1 = Bad, 2 = Good, 3 = Excellent

Problem:

No neutral or negative range.

Improved scale:

1 = Very dissatisfied, 2 = Dissatisfied, 3 = Neutral, 4 = Satisfied, 5 = Very satisfied

53. Ready Answer: Measurement Error Case

Measurement error refers to the difference between the true value of a variable and the value obtained through the measurement instrument. In questionnaire design, measurement error can occur due to unclear, biased, or poorly structured questions.

Three common errors are:

1. **Ambiguous question:** A vague question such as "Is our service good?" may be interpreted differently by respondents. It can be improved as: "How satisfied are you with the speed of our customer service response?"
2. **Leading question:** A biased question such as "How satisfied are you with our excellent service?" pushes the respondent toward a positive answer. It can be improved as: "How satisfied are you with our customer service?"
3. **Double-barreled question:** A question such as "Are you satisfied with our price and delivery?" asks two things together. It can be improved by asking separate questions on price and delivery.

Thus, properly worded questions reduce measurement error and improve accuracy of research findings.

54. Statistical Technique Selection

Sometimes Unit 9 asks you to choose a statistical technique.

How to decide:

Ask:

1. What is the dependent variable?
2. What is the independent variable?
3. Are we comparing groups or predicting relationships?

55. ANOVA

ANOVA is used to compare the mean of one variable across three or more groups.

Simple meaning:

ANOVA = compare average scores of 3 or more groups.

Example:

Customer satisfaction across:

1. Branch banking
2. Mobile banking
3. Internet banking

DV:

Customer satisfaction

IV:

Service channel with three groups

Technique:

One-way ANOVA

Hypotheses:

H0: There is no significant difference in customer satisfaction across branch banking, mobile banking, and internet banking.

H1: There is a significant difference in customer satisfaction across at least one of the service channels.

56. Regression

Regression is used when we want to study how independent variables affect a dependent variable.

Simple meaning:

Regression = predict Y using X variables.

Example:

Sales performance depends on advertising, price, store location, and discounts.

DV:

Sales performance

IVs:

Advertising, price, store location, discounts

Technique:

Multiple regression

57. Technique Selection Shortcut

Situation	Technique
Compare 2 groups	t-test
Compare 3 or more groups	ANOVA
Study relationship between two variables	Correlation
Predict one dependent variable using many independent variables	Regression
Reduce many variables into factors	Factor analysis
Group customers into segments	Cluster analysis

58. Research Report Structure

This is lower priority because it appeared mostly in older papers, but read once.

Common structure:

1. Title page
2. Executive summary/abstract
3. Introduction
4. Research problem and objectives
5. Literature review
6. Research methodology
7. Data analysis and findings
8. Discussion
9. Conclusion and recommendations
10. Limitations
11. References
12. Appendices

59. Guidelines For Research Report

A good research report should be:

- Clear
- Logical
- Well-structured
- Objective
- Supported by data
- Written in simple language
- Properly referenced
- Useful for decision-making

60. PYQ Pattern From Unit 9

Most common questions:

1. Discuss reliability and validity.
2. Explain methods to assess reliability and validity.
3. Explain goodness of measures.
4. Explain rating scales with examples.
5. Explain measurement error in questionnaire design.
6. Choose statistical technique for a case.
7. Explain operational definition.
8. Explain research report structure/guidelines.

61. Ready-Made 10-Mark Answer: Reliability And Validity

Reliability and validity are important for establishing the goodness of measures in research.

Reliability refers to the consistency of a measurement instrument. A reliable tool gives stable and consistent results when used repeatedly under similar conditions. Reliability can be assessed through test-retest reliability, parallel-form reliability, split-half reliability, internal consistency, and Cronbach's Alpha. Cronbach's Alpha of 0.70 or above is generally considered acceptable.

Validity refers to the accuracy of a measurement instrument. A valid tool measures what it is intended to measure. Validity can be assessed through content validity, criterion validity, construct validity, convergent validity, and discriminant validity.

Reliability and validity are important because they reduce measurement error and increase the trustworthiness of research findings. A tool may be reliable but not valid, but a valid tool is expected to be reliable. Thus, both are necessary for meaningful business research.

62. Ready-Made Answer: Operational Definition

Operational definition means defining an abstract concept in measurable terms. It converts broad concepts into dimensions and observable elements.

For example, consumer trust can be operationalized through dimensions such as honesty, reliability, security, and customer care. These can be measured using statements such as:

- The brand keeps its promises.
- The brand provides truthful information.
- I feel safe sharing personal information with this brand.
- The brand cares about customer problems.

Respondents can rate these items on a 5-point Likert scale from strongly disagree to strongly agree. Thus, operational definition helps convert abstract concepts into measurable variables.

63. One-Page Revision

Measurement = assigning numbers/symbols to characteristics.

Operational definition = making abstract concept measurable.

Dimension = broad part of concept.

Element = specific measurable item/question.

Reliability = consistency.

Validity = accuracy.

Goodness of measures = reliability + validity.

Cronbach's Alpha = internal consistency test.

0.70 Alpha = acceptable reliability.

Likert scale = agreement scale.

Semantic differential = opposite-word scale.

Measurement error = inaccurate measurement due to poor questions/tool.

ANOVA = compare 3 or more groups.

Regression = predict dependent variable using independent variables.

64. Most Important Unit 9 Topics To Study First

1. Reliability and validity
2. Measurement error and questionnaire correction
3. Operational definitions
4. Rating scales
5. ANOVA and regression technique selection
6. Research report only if time remains